

Design Process of Room Temperature Monitor

1. Summary

This report presents the design and application of a temperature monitoring system that can monitor temperatures within a range and send alerts when out of range. The monitor was built and tested using an alert system with a red LED and buzzer whenever temperature was out of the desired range. Materials and design were pre-determined for the monitor, but the focus is on the results that its utility compared to normal thermometers. The system's ability to alert quickly allows users to act and control temperature whenever necessary. While the work of the monitor is significant, it shows a more innovative alternative to regular temperature measuring methods. Furthermore, even technologically advanced thermometers are already available and used in many homes and businesses and this report shows the usefulness of these systems.

2. Introduction

In our daily lives, temperature plays a massive role in everything such as cooking, outfits, and freezing. Most of the time, the temperature users feel room temperature in indoor environments such as a home or office but even then, temperature fluctuations affect people's comfort and energy. The report explores the design process of a temperature monitoring system, ranging from concept to application and aims to prototype a monitor that measures room temperature accurately, provide real time response through LED displays, and alert people of significant temperature changes. The purpose of the report is to document the design process of the room temperature monitoring system throughout drafting, material selection, prototype, and testing. By documenting the process, it serves as a way for future temperature monitors or similar projects.

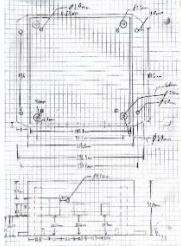
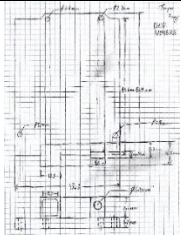
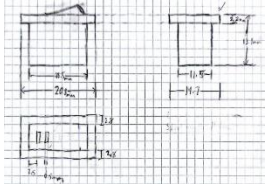
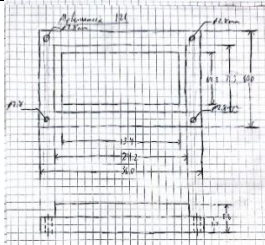
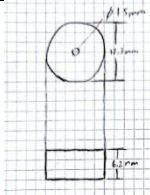
3. Design Elements

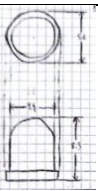
a.

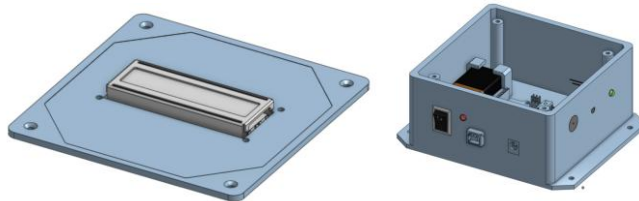
1. Enclosure
2. Arduino Uno board
3. 9V battery
4. 9V battery snap
5. USB cable
6. Breadboard 400 points
7. Solid core jumper wires
8. Stranded jumper wires (red and black)
9. 22 AWG solid/stranded wires
10. Temperature sensor
11. Alphanumeric I2C LCD (16x2 characters)
12. LEDs (red and green)
13. Piezo capsule (buzzer) [PKM17EPP-4001-B0]
14. Resistors
15. 2-way way switch

Figure 1: Diagram of the component list of Room Temperature Monitor

b. Table 1: Precision Measurements of the monitor's components

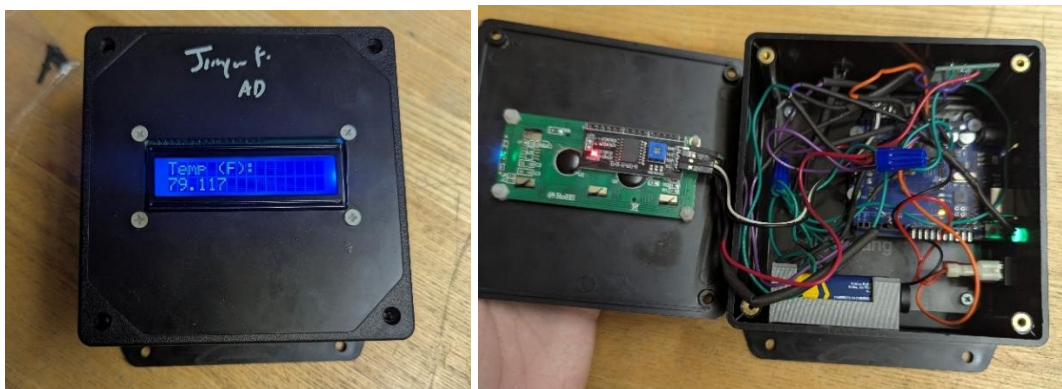
Item	Sketch Identification	W [mm]	L [mm]	H [mm]	Diameter [mm]
ABS Enclosure	 (Top + Front View)	110.1	146.4	57.0	N/A
Arduino Board	 (Top View)	53.2	68.4	12.8	N/A
Switch	 (All Views)	14.7	20.8	13.1	N/A
LCD 2x16	 (Top and Front View)	36	80	8.6	N/A
Buzzer		N/A	N/A	6.2	12.3

	(Top and Front View)				
Temperature Sensor	 (Top and Front View)	N/A	N/A	4.6	3.5
LED	 (Top and Front View)	N/A	N/A	8.3	4.9



c.

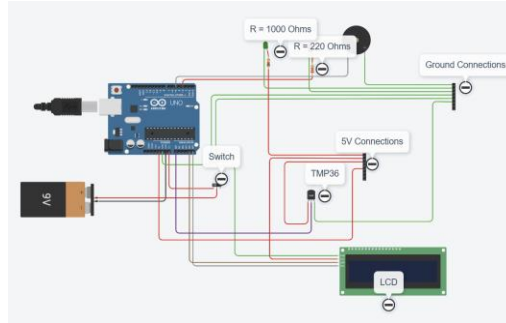
Figure 2: Completed CAD Assemblies of the temperature monitor's lid and box



d.

Figure 3: Top Views of Prototypes with lid (Left) and without lid (Right)

- e. The purpose of the Arduino board is to act as the control of all the electronic components in the device. Firstly, it receives analog voltage signals and converts them to discrete voltage signals which is used by the TMP36 and LCD to read and display temperature.



f.

Figure 4: Final Wiring Diagram of temperature monitor circuit

- g. The wire gauge used in the monitor box are both the 22 AWG and jumper wires. Jumper wires are used in the LCD to connect the signals SDA and SCL to the Arduino board which is only used because the voltage drop throughout the wire should not significantly affect transmitted signals from the Arduino to LCD. Everything else is connected using 22 AWG wires because it has less voltage drop and a more secure connection. The resistors used for the Green LED is 1000 Ω and the Red LED is 220 Ω excluding internal resistance in the wires or the LEDs. The operating current for the green LED is around 5.2 mA and the red LED is around 13.9 mA, respectively. This makes sense because Kirchoff's Voltage Law (KVL) states voltage around a closed loop is 0 and inverse relationship between current and resistance so a higher resistance gives a lower current and vice versa.
- h. A 9V battery is used because the battery provides enough voltage for the Arduino's 6 to 20 V range. The battery has a charging capacity of 500 mAh which is on the lower side of batteries, which would make this a bad choice as it would not last a long time for the purpose of monitoring temperature over a period. The Arduino has a power jack which can be used as an external power source.

```

1 // LCD setup & Pins
2 #include <LiquidCrystal_I2C.h> // Include LCD library
3 LiquidCrystal_I2C lcd(0x27, 16, 2); // I2C address 0x27, 16x2 display
4
5 const int tempPin = A0; // TMP36
6 const int redLEDPin = A5; // Red LED
7 const int buzzerPin = A7; // Buzzer
8
9 void setup() {
10   pinMode(redLEDPin, OUTPUT);
11   pinMode(buzzerPin, OUTPUT);
12   Serial.begin(9600); // Start serial monitor
13   lcd.init(); // Initialize LCD
14   lcd.backlight(); // Turn on backlight
15 }
16
17 void loop() {
18   int sensorValue = analogRead(tempPin); // Read TMP36
19   float voltage = sensorValue * (5.0 / 1024.0); // Convert to volts
20   float temperatureC = (voltage - 0.5) * 100; // 100mV formula
21   float temperatureF = (temperatureC * 9.0 / 5.0) + 32.0; // To F
22   Serial.print("Temp (F): "); // Show Temperature
23   Serial.println(temperatureF);
24   lcd.setCursor(0, 0);
25   lcd.print("Temp (F):");
26   lcd.setCursor(0, 1);
27   lcd.print(temperatureF);
28
29   if (temperatureF < 70 || temperatureF > 80) {
30     digitalWrite(redLEDPin, HIGH); // Turn on red LED and buzzer
31     tone(buzzerPin, 1000);
32     delay(1000);
33     digitalWrite(redLEDPin, LOW); // Turn off red LED and Buzzer
34     noTone(buzzerPin);
35     delay(1000);
36   } else {
37     digitalWrite(redLEDPin, LOW); // Keeps buzzer and LED off
38     noTone(buzzerPin);
39     delay(1000);
40   }
41 }

```

i.

Figure 5: Arduino Code for temperature monitoring

- j. Specifications

By using equation 1 with a battery capacity of 500 mAh and current drawn of 14.06 mA,

$$\text{Charging Capacity} = \frac{\text{Battery Capacity (mAh)}}{\text{Drawn Current (mA)}} = \frac{500 \text{ mAh}}{14.06 \text{ mA}} = 35.56 \text{ hrs} \quad (1)$$

Specifications: Voltage of Power Supply is 9.45 V, Operating Circuit of the circuit is 5.41 V, Total Current Drawn is 14.06 mA, Battery Operating Time is 35.56 hours.

The TMP36 in our temperature monitor has an operating range from -55 °C to 150 °C so a very comfortable temperature range is from 0 °C to 100 °C. The idea of using a 1 kΩ resistor for the green LED and 200 Ω resistor for the red LED is because of the desired brightness level. According to KVL, resistance and current have an inverse relationship in a closed path of 0 voltage. Therefore, the green LED receives less current and red LED receives more current causing the green LED to be less bright than the red LED because the red LED's purpose is to send a signal when it is out of the temperature range while the green LED is used to indicate when the monitor is on.

4. Evaluation of Results

The construction and application of the temperature monitoring system has achieved all the design objectives described by accurately measuring a temperature range, red LED flashes, and buzzer alerts. The outcome of this project is to develop a compact, user-friendly, and functional temperature monitor. Compared to standard thermometers, this monitor has an additional alert system for temperatures not within the desired range. However, this project also has limitations. The battery holder is restrained in a place inside the box with a bunch of wires which makes it hard to organize and difficult to remove the battery from its holder. Additionally, the battery life is not good for this system which would cause trouble replacing it every time as an internal battery source. Additionally, the prototype is very fragile due to the materials used. Future improvements can be an easier way to remove the battery or compact the box to be more space efficient, or materials used to be sturdier. The usefulness of the work is shown in its ability to detect temperatures in various scenarios and send alerts, buzzer noise and flashing red LED, due to the alert system that makes it more convenient for users than a regular thermometer.